

Rakesh Jha

Platform & Cloud Operations Engineer

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EDUCATION

Guru Jambheshwar University of Science & Technology (GJUS&T)

Bachelor of Technology in Information Technology

Hisar, India

2021 – 2025

Core Coursework: Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN), Theory of Computation (TOC), Systems Design, Data Structures and Algorithms.

SKILLS SUMMARY

- **Cloud & Platforms:** Amazon Web Services (AWS Architecture, VPC, EC2, IAM, Route53, Security Groups, NACLs, S3), Linux OS (Ubuntu Internals & Bash Scripting).
- **Containerization & Orchestration:** Kubernetes (K8s Clusters, Pod Networking, Service Discovery, Cluster Architecture), Docker, Docker Compose.
- **Automation & CI/CD:** Jenkins (Pipelines-as-Code), ArgoCD GitOps, GitHub Actions, Terraform (IaC), Ansible, Git.
- **Languages & Databases:** Python, Java, Advanced Bash/Shell Scripting, SQL (Relational Schemas).
- **Observability & Tools:** Prometheus, Grafana, JIRA, Dynamic Port Mapping, Uptime Metrics, Distributed Systems Logic.

TECHNICAL EXPERIENCE

Independent DevOps & Systems Engineering Practitioner

Multi-Environment Infrastructure & Automation Track

Remote

April 2026 – Present

- **Hosting Compute Design:** Provisioned high-availability cloud layouts on AWS using Terraform modules, configuring Nginx reverse proxies for secure web routing and using PM2 cluster supervision to maintain reliable application self-healing and uptime.
- **Release Engineering:** Implemented automated multi-environment CI/CD release tracks with Jenkins and GitHub Actions, incorporating automated unit testing, syntax validation check gates, and ArgoCD GitOps loops to increase deployment velocity.
- **Platform Telemetry:** Programmed modular Python and Bash telemetry daemons anchored in Linux networking internals to capture live compute resource allocations, parsing network connection states to dynamically audit production metrics.

PROJECTS

Highly Available Web Infrastructure & Cloud Governance Framework (AWS, Bash)

AWS VPC, ASG, ALB Project

- Engineered a custom **AWS VPC** spanning across multiple Availability Zones (**Multi-AZ**) implementing isolated public and private subnet topologies to eliminate single points of failure.
- Configured robust perimeter network defense using tightly locked **Security Groups** and Network Access Control Lists (**NACLs**) to completely isolate backend application ec2 targets from public exposure.
- Deployed an **Application Load Balancer (ALB)** integrated with **Route53** for public traffic routing, backed by an **Auto Scaling Group (ASG)** to dynamically scale compute layers under high traffic strain.
- Built a secure network bridge utilizing a **Bastion Host (Jump Box)** with granular **IAM Identity policies** to handle backend cluster administration without public endpoints.
- Developed a companion modular **Bash/Shell governance script** using the AWS CLI to automate active cloud resource monitoring, auditing dangling instances and tracking cost footprints over crontab schedules.

Automated GitOps Engine & CI/CD Release Pipeline

Jenkins, ArgoCD, GitHub Actions

- Built a high-throughput, standardized continuous integration framework capable of handling multi-stage syntax compilation, structural code linting, and automated testing gates efficiently.
- Implemented declarative configurations and state reconciliation loops to eliminate manual infrastructure drift and ensure safe, repeatable promotions.

Multi-Container Platform Isolation & Microservices Network

Docker, Kubernetes, Terraform

- Containerized a decoupled software application ecosystem, constructing optimized multi-stage Dockerfiles to minimize container image footprints and secure deployment boundaries.
- Designed isolated container networking topologies within a **Kubernetes** cluster using custom virtual network bridges to facilitate private cross-tier **Service Discovery** while keeping public host interfaces closed.

VOLUNTEER EXPERIENCE

Speakathon Club under Training and Placement Cell, GJUS&T

Club Coordinator

Hisar, India

July 2023 – July 2024

- Led a cross-functional student team to design, organize, and execute communication bootcamps, technical presentation sessions, group debates, and simulated mock interviews for 200+ engineering candidates.
- Boosted candidate placement readiness and professional confidence by establishing systematic workshops focused on interview mechanics, public speaking, and team collaboration.